

Program: Diploma in Automation and Robotics	
Course Code : 4338	Course Title: Mechanics of Robotics Lab
Semester : 4	Credits: 1.5
Course Category: Program Core	
Periods per week: 3 (L: 0 T: 0 P: 3)	Periods per semester: 30

Course Objectives:

- Study and demonstrate the kinematic behavior of mechanical linkages
- Study the working of different mechanical drives, vibrations and balancing of masses
- Perform kinematic and dynamic analysis with simulation

Course Prerequisites:

Topic	Course Code	Course Name	Semester
Basic knowledge of kinematics of machine elements		Kinematics of machines	4

Course Outcomes:

On completion of the course, the student will be able to:

COn	Description	Duration (Hours)	Cognitive Level
CO1	Demonstrate the kinematic behavior of linkages and Cams	6	Applying
CO2	Study the working of mechanical drives	9	Applying
CO3	Demonstrate the vibration of mechanical elements and to study the balancing of forces	3	Applying
CO4	Perform kinematic and dynamic analyses with simulation.	6	Applying
	Series Test	6	

CO – PO Mapping

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	2			3			
CO2	2			3			
CO3	2			3			
CO4	2			3			

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline (Perform minimum Eight Experiments)

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	The study kinematic behavior of linkages and cam		
M1.01	Kinematics of Four Bar, Slider Crank and their inversions	3	Applying
M1.02	Kinematics of single and double universal joints and Cam profile drawing	3	Applying
Experiments 1. Kinematics of Four Bar, Slider Crank and their inversions 2. Kinematics of single and double universal joints. 3. Cam - Cam profile drawing, Motion curves and study of jump phenomenon			
CO2	Study the working of mechanical drives		
M2.01	Study the working of belt, chain and rope drives	3	Applying
M2.02	Experimental study of velocity ratios of simple, compound, Epicyclic and differential gear trains.	3	Applying
Experiments 4. Study the working of belt, chain and rope drives 5. Study gear parameters. 6. Experimental study of velocity ratios of simple, compound, Epicyclic and differential gear trains			
	Series Test – I	3	

CO3	Demonstrate the vibration of mechanical elements and to study the balancing of forces		
M3.01	Study of gyroscopic effect and couple.	3	Applying
M3.02	Study of longitudinal and torsional vibration	3	Applying
M3.03	Whirling of shafts	3	Applying
M3.04	Balancing of masses	3	Applying
Experiments 7. Motorized gyroscope – Study of gyroscopic effect and couple. 8. Single degree of freedom Spring Mass System – Determination of natural Frequency 9. Study of torsional vibration in shaft (single rotor and two rotor system) 10. To determine whirling speed of the shaft with concentrate loading 11. Understand and verify the fundamental laws of static & dynamic balancing 12. Study balancing of rotating and reciprocating masses.			
CO4	Perform kinematic and dynamic analyses with simulation.		
M4.01	Perform kinematic and dynamic analyses with simulation.	3	Applying
M4.02	Perform inverse kinematic analyses with simulation.	3	Applying
Experiments 13. Forward kinematics and validation using a software (Robo Analyser/Matlab or any other free software tool) 14. Inverse kinematics of an industrial robot and validation using any open-source software			
	Series Test– 2	3	

Text / Reference

T/R	Book Title / Author
R1	Theory of Machines, S.S.Rattan , Tata Mc-Graw Hill.
R2	Mechanical Vibrations and Noise Engineering, A. G. Ambekar, Prentice Hall of India.
R3	Robotics Engineering : R. Klafter, PHI